Bryan Tseng

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Education

Johns Hopkins University School of Medicine, Baltimore, MD — Aug 2024 – Present
Ph.D. in Biomedical Engineering — Advisor: Dr. Adam S. Charles
Activities: Treasurer, BME Ph.D. Student Council

Boston University College of Arts and Sciences, Boston, MA — Sep 2016 – May 2020
B.A. in Mathematics (Statistics) and Biology (CMG)
Minor in Economics
Awards: Dean's List (all semesters), UROP Research Award (Fall 2019, Spring 2020), *magna cum laude*

Publications

Peer-Reviewed

1. Guay C, Agrawal U, Tseng B, Gallo S, Schreier D, Brown E (2025). Clinical Electroencephalography for Anesthesiologists and Intensivists Part II: Physiologic Signatures and Active Management. *Anesthesiology*.
2. Breault M, Orguc S, Kwon O, Kang G, Tseng B, Gallo S, Schreier D, Brown E (2024). Anesthetics As Treatments For Depression: Clinical Insights and Underlying Mechanisms. *Annual Review of Neuroscience*.
3. Subramanian S, Tseng B, Carmen M, Goodman A, Dahl D, Barbieri R, Brown E (2024). Monitoring Surgical Nociception Using Multimodal Physiologic Markers. *PNAS*.
4. Subramanian S, Tseng B, Barbieri R, Brown E (2022). An unsupervised automated paradigm for artifact removal from electrodermal activity in an uncontrolled clinical setting. *Physiological Measurement*.
5. Marshall A, Joyce C, Tseng B, et al. (2021). Changes in Pain Self-Efficacy, Coping Skills and Fear Avoidance Beliefs in a Randomized Controlled Trial ... *Pain Medicine*.
6. Nojiri T, Chen CY, Kim DM, Tseng B, et al. (2019). Establishment of perpendicular protrusion of type I collagen on TiO_2 nanotube surface ... *Journal of Nanobiotechnology*.

Conference Papers

1. Tseng B, Subramanian S, Barbieri R, Brown E (2022). Tonic Electrodermal Activity is a Robust Marker of Psychological and Physiological Changes ... *IEEE EMBC*.
2. Subramanian S, Tseng B, Barbieri R, Brown E (2021). Unsupervised Machine Learning Methods for Artifact Removal in Electrodermal Activity *IEEE EMBC*.

Conference Abstracts

1. Tseng B, Subramanian S, Purdon P, Barbieri R, Brown E (2021). Prediction of unconsciousness using clinically available markers. *IEEE EMBC*.
2. Tseng B, Subramanian S, Purdon P, Barbieri R, Brown E (2021). Multimodal brain-heart analysis reveals subject-specific dynamics during propofol anesthesia *OHBM*.

Provisional Patent

1. Combinations of Anesthetics as Therapies for Treatment-Resistant Depression (MIT 25342)

Research Experience

Johns Hopkins University, Baltimore, MD — Aug 2020 – May 2024
Graduate Research Assistant – Dr. Adam S. Charles

- **Research topic:** Developing **adaptive stimulation algorithms** for brain-computer interfaces by learning interpretable models of **neural dynamics** and predictive models of movement.
- Designed reaching and head-turn behavioral paradigms to study the effect of **microstimulation** in the deep cerebellar nuclei using **Neuropixel Read Write (NPRW) prototype probes**
  - Assisted with the non-human primate neural recording
- Built **end-to-end analysis pipelines for spike sorting large-scale electrophysiology data** using SpikeInterface for **Utah arrays**, NPRW probes, and **DeepLabCut** annotated video kinematics data
  - Neural recording analysis pipeline included artifact correction of stimulation artifact, spike detection, and firing rate estimation

- Analyzed the parameter space of NPRW microstimulation and its corresponding effects on behavior kinematics and evoked neural responses (cortical and deep brain)
- Analyzed behaviorally relevant subspaces of neural dynamics in the deep cerebellar nuclei
- Deployed analysis pipelines on the high-performance computing cluster provided by Center of Imaging Science

Massachusetts Institute of Technology / MGH, Cambridge, MA — Aug 2020 – May 2024
Neuroscience Statistics Research Laboratory (NSRL), Massachusetts General Hospital
Technical Associate — Dr. Emery N. Brown

- Implemented state-space multitaper spectral analysis on the electroencephalography (EEG) to track patient's level of consciousness during general anesthesia
  - Assisted in integrating the online EEG metrics into an existing closed-loop anesthesia delivery system
- Implemented point-process and state-space methods to track patient's nociceptive state during general anesthesia
  - Derived from a history-dependent Inverse Gaussian model on the electrocardiogram (EKG) and electrodermal activity (EDA), reaching accuracies of up to 75%
- Investigated the neurophysiological mechanisms underlying the antidepressant effects of anesthetics
- Investigated temporal dynamics of brain activity, autonomics, and behavior due to propofol administration
- Standardized the lab database and analysis pipeline and integration with the MIT OpenMind computing cluster
- Fascilitated the collaboration between NSRL and the Barbieri lab at Politecnico di Milano

Clinical Responsibilities — Massachusetts General Hospital

- Led and initiated passive data collection efforts of physiological signals in the ICU, operating room, and electroconvulsive therapy units
  - Physiological signals from 100+ patients (EEG, EKG, EDA, photoplethysmography)
- Performed artifact correction and peak extraction of physiological signals using machine learning methods
  - Tested isolation forest, K-nearest neighbor distance 1-class support vector machine methods to remove cautery artifact in data collected during surgery
- Investigated the clipping and scaling issues of EEG signals from the Masimo Sedline Root device
- Managed IRB protocols and data use agreements at MGH

Boston Medical Center, Boston, MA — Jun 2019 – Aug 2020
Research Assistant — Dr. Robert Saper and Eric Roseen

- Investigated psychometric effects of yoga and physical therapy on adults with chronic lower back pain
  - Assessed within and between-group change scores of covariates from the Back to Health study in SAS
  - Implemented stochastic regression imputation of missing data
- Investigated attitudes and beliefs of opioid use disorder patients on non-pharmacologic treatment
- Performed Fisher exact analysis on the effects of nonpharmacologic treatment relative to pain medication usage
- Funded by the Undergraduate Research Opportunities Program (UROP) (Fall 2019 and Spring 2020)

Leadership Experience

Allocation Board of Boston University, Boston, MA — Sep 2017 – May 2020
Chair

- Worked directly with the Dean of Students to allocate over $700,000 annually to 300 clubs
- Led weekly meetings to approve funding requests from student organizations
- Worked with club presidents and treasurers to ensure the proper use of funding
- Rewrote the board constitution, handbook, performed audits, and helped design a new recruitment process
- Facilitated the transition to a new finance and event planning platform through workshops and advertisement
- Interviewed and led training processes for new members
- Helped to coordinate large campus-wide events such as the first BU spring concert

Volunteer Experience

- Peer mentor at Boston University Department of Mathematics and Statistics — Sep 2019 – May 2020
  - One-on-one mentorship for Freshman students, hosted department-sponsored events
- Big Brothers and Big Sisters of Mass Bay — Sep 2017 – May 2019
  - Mentorship of youth in the Greater Boston Area in a biweekly camp